Industrial Internship Report on

Air Quality Monitoring System

Prepared by

Sangeetha R]

Executive Summary
This report provides details of the Industrial Internship provided by upskill Campus and The IoT Academy in collaboration with Industrial Partner UniConverge Technologies Pvt Ltd (UCT). This internship was focused on a project/problem statement provided by UCT. I had to finish the project including the report in 6 weeks' time. My project was an IoT-based Air Quality Monitoring System. The objective of this project was to monitor environmental conditions such as temperature, humidity, gas levels, and Air Quality Index (AQI) using sensors and a microcontroller. The system collects data from sensors and sends it to a cloud platform for real-time monitoring and visualization. This project helped me understand how IoT systems work in real-world applications and how environmental data can be used for safety and awareness. This internship gave me a very good opportunity to get exposure to Industrial problems and design/implement solution for that. It was an overall great experience to have this internship.

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1 Preface

This report encapsulates my journey during the 6-week internship conducted under Upskill Campus in collaboration with UniConverge Technologies Pvt. Ltd.. More than just an academic requirement, this internship became a transformative experience that reshaped my understanding of technology, problem-solving, and my own potential.

In today's rapidly evolving technological landscape, theoretical knowledge alone is not sufficient. Practical exposure is essential to truly understand how concepts translate into real-world applications. This internship served as that crucial bridge, allowing me to move beyond passive learning and actively engage in building a complete IoT solution from the ground up.

The assigned project, Air Quality Monitoring System, introduced me to a problem that is both technically relevant and socially significant. It was not merely about implementing sensors and writing code, but about understanding how such systems can contribute to monitoring environmental health and improving quality of life.

The structure of the program was thoughtfully designed, progressing from foundational concepts to system design, implementation, data analysis, and final documentation. Each phase demanded not only technical effort but also critical thinking and adaptability.

Throughout this journey, I encountered several challenges—ranging from understanding sensor behavior to dealing with simulation limitations and unsuccessful attempts at implementing tools like Node-RED. These moments of difficulty were not setbacks, but turning points. They pushed me to think deeper, troubleshoot independently, and develop resilience.

One of the most significant realizations during this internship was that true learning happens outside comfort zones. Every error, every failed attempt, and every moment of confusion contributed to a clearer and stronger understanding of the subject. Over time, I noticed a shift—not just in my technical skills, but in my confidence and approach towards problem-solving.

This internship has played a vital role in my growth, both technically and personally. I have developed a structured way of thinking, improved my ability to analyze problems, and gained confidence in handling real-world projects.

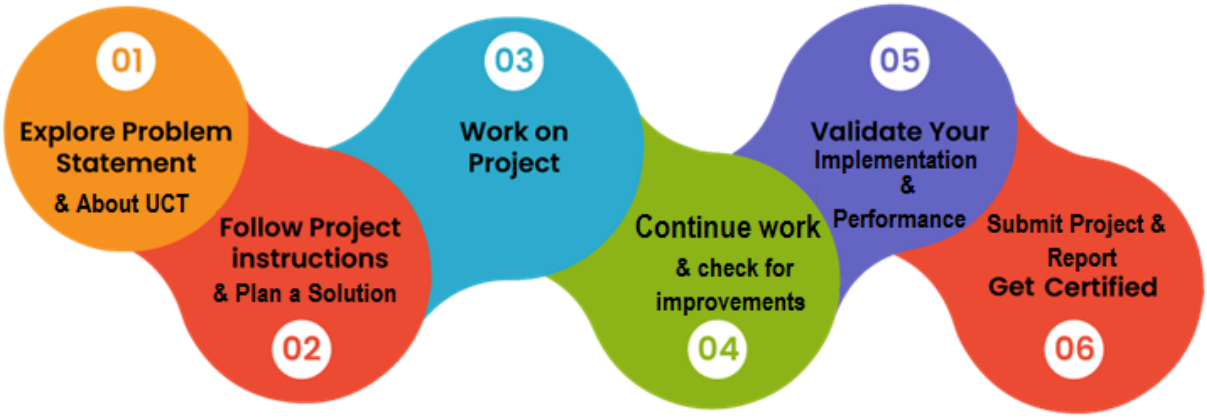
II would like to express my sincere gratitude to the team of Upskill Campus and UniConverge Technologies Pvt. Ltd. for providing me with this valuable opportunity to learn and grow.

I would also like to thank Ms. Prabha Singh (HR Manager) for her support and coordination during the internship.

I am grateful to all the mentors and coordinators who guided us throughout the program and helped in understanding the concepts clearly.

I would also like to thank my friends and family for their constant support and encouragement during this internship.

To my juniors and peers, I would like to share this—do not measure your progress only by results. Focus on the effort, the learning, and the persistence you build along the way. Challenges are not obstacles; they are opportunities to grow. Stay consistent, stay curious, and trust your ability to learn and improve.



2 Introduction

2.1 About UniConverge Technologies Pvt Ltd

A company established in 2013 and working in Digital Transformation domain and providing Industrial solutions with prime focus on sustainability and RoI.

For developing its products and solutions it is leveraging various **Cutting Edge Technologies e.g. Internet of Things (IoT), Cyber Security, Cloud computing (AWS, Azure), Machine Learning, Communication Technologies (4G/5G/LoRaWAN), Java Full Stack, Python, Front end** etc.

i. UCT IoT Platform (Insight)

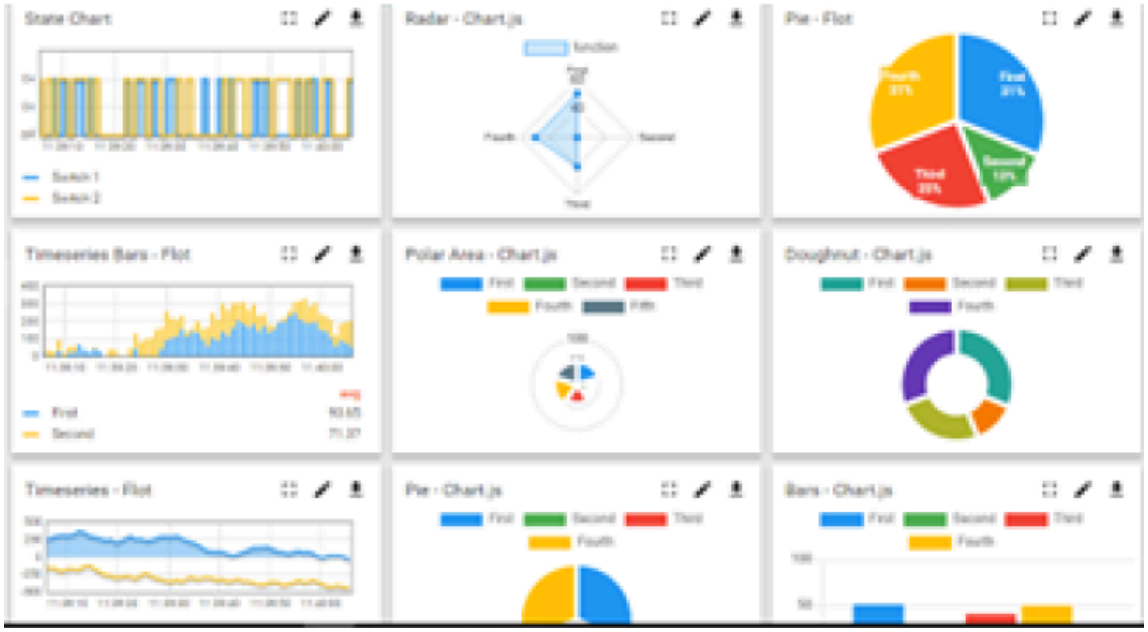
UCT Insight is an IOT platform designed for quick deployment of IOT applications on the same time providing valuable "insight" for your process/business. It has been built in Java for backend and ReactJS for Front end. It has support for MySQL and various NoSql Databases.

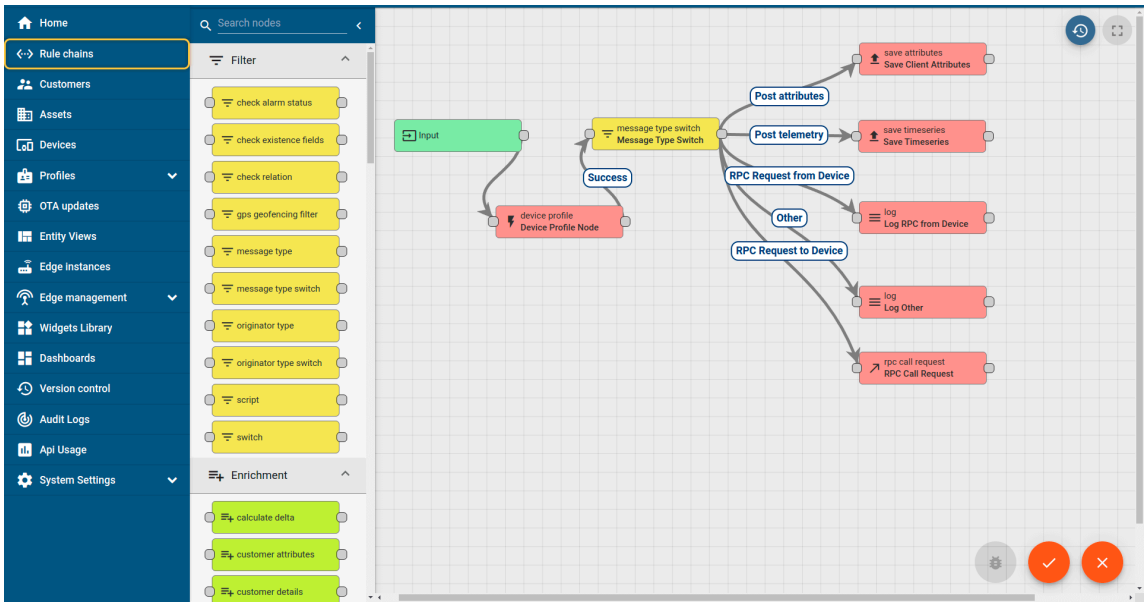
- It enables device connectivity via industry standard IoT protocols - MQTT, CoAP, HTTP, Modbus TCP, OPC UA

- It supports both cloud and on-premises deployments.

It has features to

• Build Your own dashboard
• Analytics and Reporting
• Alert and Notification
• Integration with third party application(Power BI, SAP, ERP)
• Rule Engine





ii. Smart Factory Platform (FACTORY WATCH)

Factory watch is a platform for smart factory needs.

It provides Users/ Factory

- with a scalable solution for their Production and asset monitoring
- OEE and predictive maintenance solution scaling up to digital twin for your assets.
- to unleased the true potential of the data that their machines are generating and helps to identify the KPIs and also improve them.
- A modular architecture that allows users to choose the service that they what to start and then can scale to more complex solutions as per their demands.

Its unique SaaS model helps users to save time, cost and money.

Machine	Operator	Work Order ID	Job ID	Job Performance	Job Progress		Output		Rejection	Time (mins)				Job Status	End Customer
					Start Time	End Time	Planned	Actual		Setup	Pred	Downtime	Idle		
CNC_S7_81	Operator 1	WO0405200001	4168	58%	10:30 AM		55	41	0	80	215	0	45	In Progress	i
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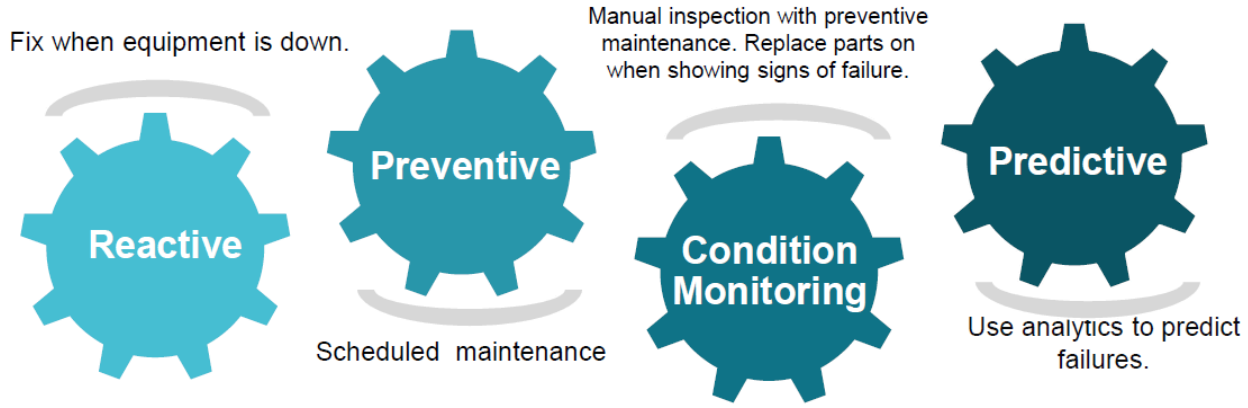
iii. LoRaWAN based Solution

UCT is one of the early adopters of LoRAWAN teschnology and providing solution in Agritech, Smart cities, Industrial Monitoring, Smart Street Light, Smart Water/ Gas/ Electricity metering solutions etc.

iv. Predictive Maintenance

UCT is providing Industrial Machine health monitoring and Predictive maintenance solution leveraging Embedded system, Industrial IoT and Machine Learning Technologies by finding Remaining useful life time of various Machines used in production process.



2.2 About upskill Campus (USC)

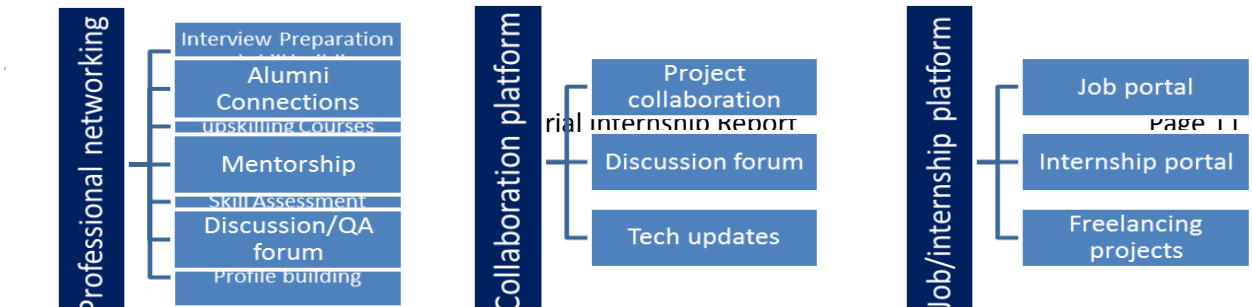
upskill Campus along with The IoT Academy and in association with Uniconverge technologies has facilitated the smooth execution of the complete internship process.

USC is a career development platform that delivers **personalized executive coaching** in a more affordable, scalable and measurable way.

Seeing need of upskilling in self paced m additional support services e.g. Internship, projec ndustry experts, Career growth Services

hi lcampus.com/

upSkill Campus aiming to upskill 1 million ext 5 year



2.3 The IoT Academy

The IoT academy is EdTech Division of UCT that is running long executive certification programs in collaboration with EICT Academy, IITK, IITR and IITG in multiple domains.

2.4 Objectives of this Internship program

The objective for this internship program was to

- ☛ get practical experience of working in the industry.
- ☛ to solve real world problems.
- ☛ to have improved job prospects.
- ☛ to have Improved understanding of our field and its applications.
- ☛ to have Personal growth like better communication and problem solving.

2.5 Reference

[1]ThingSpeak Documentation

[2] Wokwi Official Simulator

[3] ESP32 Technical Documentation

2.6 Glossary

Terms	Acronym
IoT	Internet of Things
ESP32	Microcontroller with WiFi
AQI	Air Quality Index
API	Application Programming Interface
MQTT	Message Queuing Telemetry Transport

3 Problem Statement

The problem statement assigned for this internship is to design an IoT-based Air Quality Monitoring System.

The system should monitor environmental conditions such as temperature, humidity, and gas levels, and determine the air quality index. The collected data should be transmitted to a cloud platform for real-time monitoring and analysis.

The goal is to develop a smart monitoring solution that can be used in homes, offices, and industries to ensure a safe and healthy environment.

4 Existing and Proposed solution

Existing Solutions

Existing air quality monitoring systems:

- Use expensive industrial sensors
- Require complex hardware setup
- Lack real-time cloud-based monitoring in low-cost setups

Limitations

- High cost
- Limited accessibility
- Complex configuration
- Not suitable for beginners or small-scale applications

4.1 Code submission (Github link)

https://github.com/thegreatsangeethanibli/upskillcampus/blob/main/AirQualityMonitoringSystem.ino

4.2 Report submission (Github link) :

https://github.com/thegreatsangeethanibli/upskillcampus/blob/main/AirQualityMonitoringSystem_Sangeetha_USC_UCT%20(1).pdf

Demo Video:

https://drive.google.com/file/d/11YK9ulbbEblZM3bsQVKC-wU909iPjmsK/view?usp=drivesdk

4.3 Proposed Solution

The proposed system is a low-cost IoT-based air quality monitoring system using:

- ESP32 microcontroller
- MQ-2 gas sensor
- DHT22 sensor
- Cloud platform for visualisation

4.4 Value Addition

- Low-cost implementation
- Real-time monitoring using cloud
- Simulation-based development (no hardware required)
- Easy to deploy and scalable

5 Proposed Design/ Model

- The system follows a structured IoT architecture:
- Design Flow
- Sensors collect environmental data
- ESP32 processes the data
- WiFi sends data to cloud
- Cloud platform visualizes data

5.1 High Level Diagram

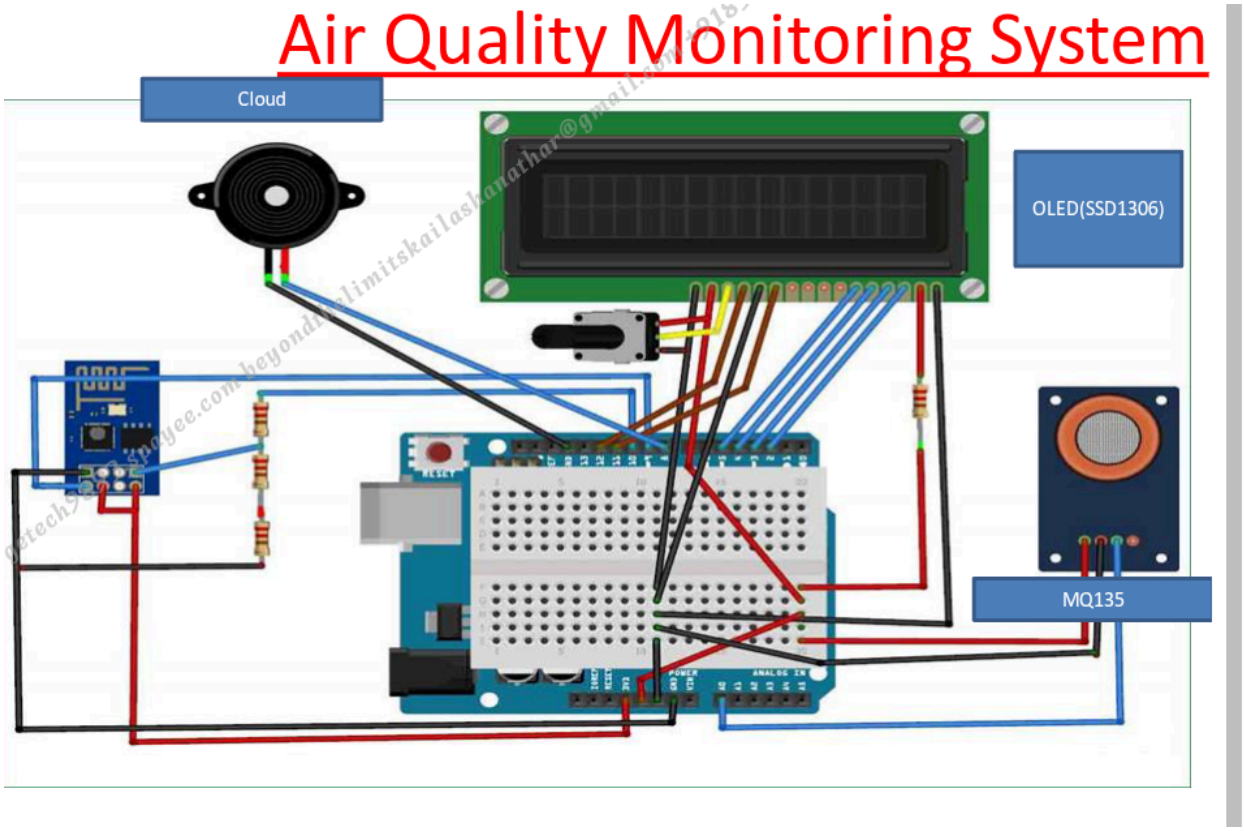
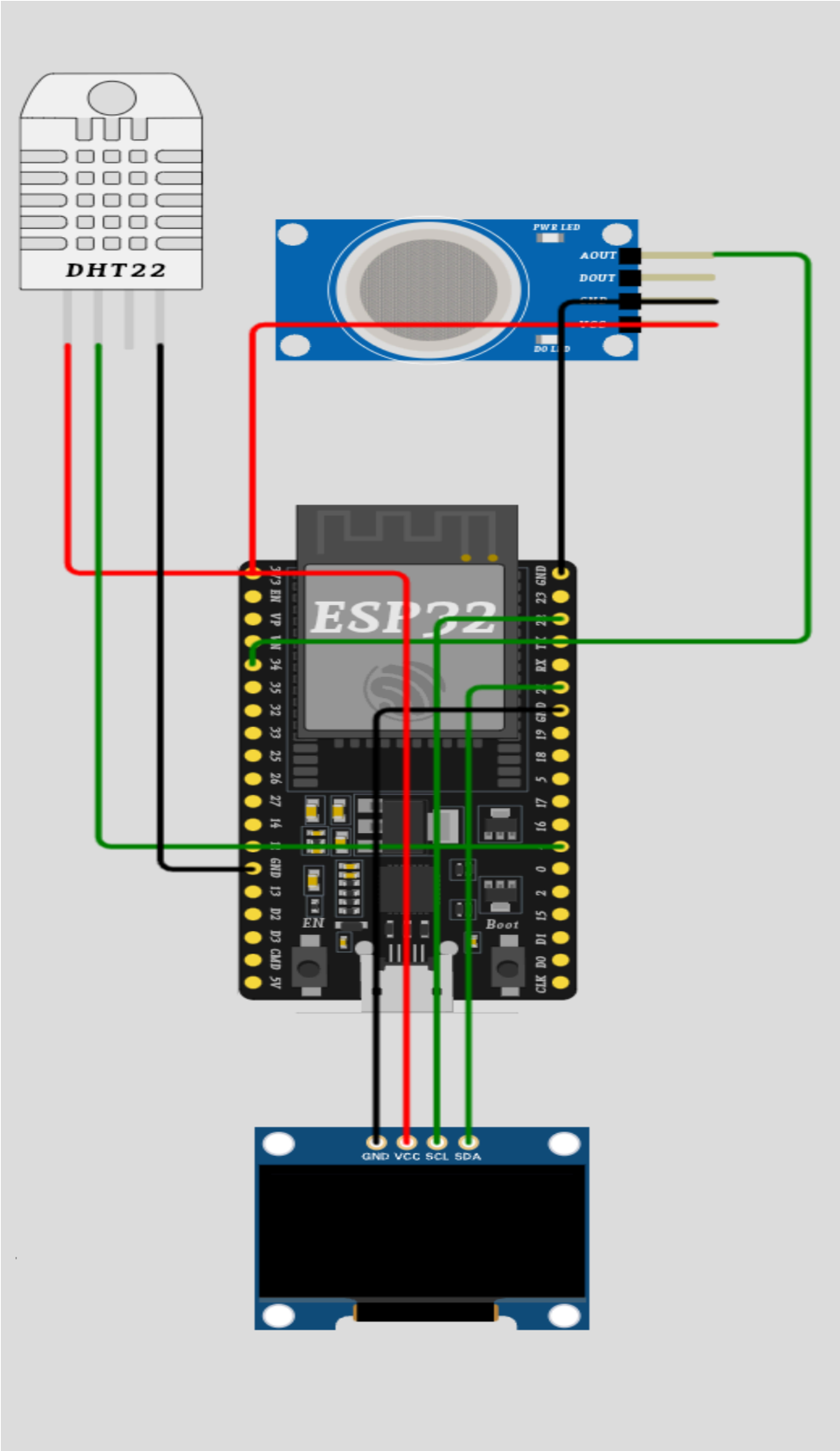
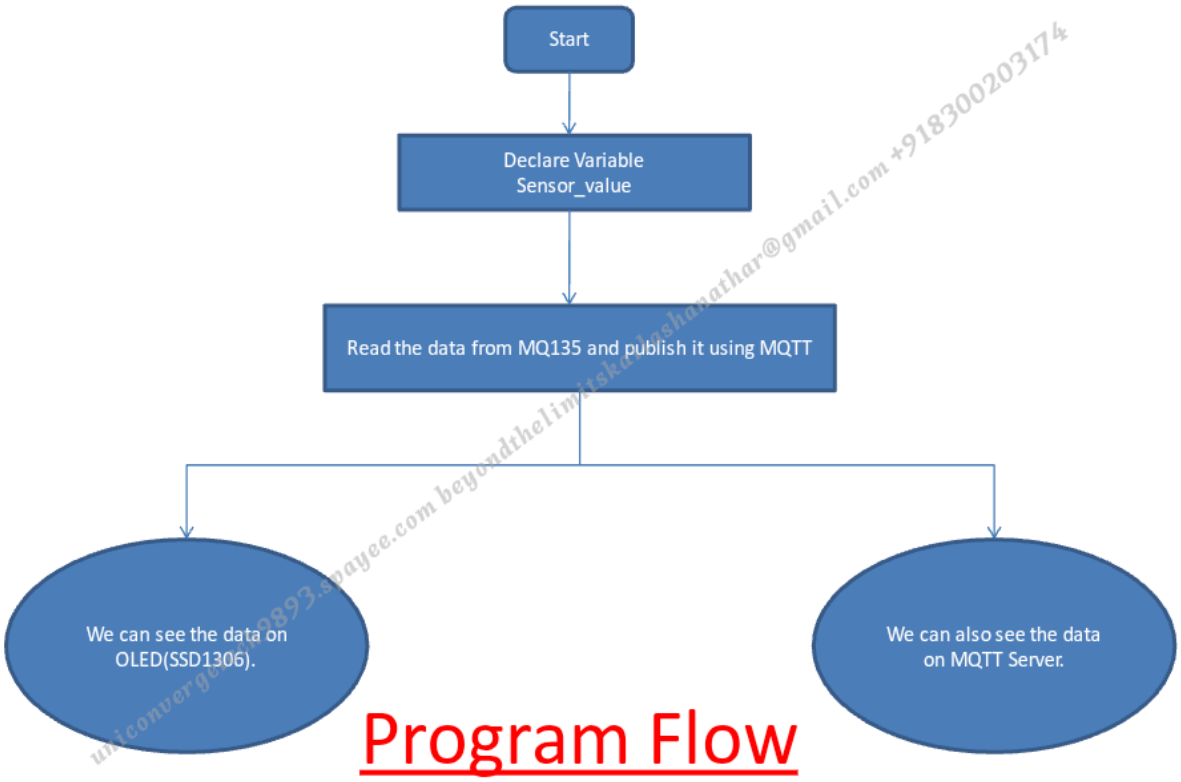


Figure 1: HIGH LEVEL DIAGRAM OF THE SYSTEM

5.2 Low Level Diagram (if applicable)


DHT22
PWR LED
AOUT
DOUT
ESP32
EN
Boot
GND VCC SCL SDA

5.3 Interfaces



Data Flow

Sensor Data → ESP32 → HTTP Request → Cloud Platform

Protocols Used

HTTP (for data transmission)

WiFi communication

Flow Description

- Sensor reads data
- ESP32 processes values
- Data formatted into URL
- Sent to cloud server
- Data visualized in graphs

6 Performance Test

Constraints Identified

- Network latency
- Sensor accuracy
- Data transmission delay
- Simulation limitations

Handling Constraints

- Maintained 15-second delay for stable transmission
- Used reliable cloud platform
- Simplified data processing

6.1 Test Plan/ Test Cases

- The first test case is Sensor Reading, where the system reads temperature and humidity values from the DHT22 sensor. The expected result is that accurate and consistent temperature and humidity values are obtained.
- The second test case is Gas Detection, where the MQ-2 sensor detects the presence of gases or smoke. The expected result is that the gas sensor value changes according to the surrounding environment conditions.
- The third test case is Cloud Upload, where the ESP32 sends the collected sensor data to the cloud platform. The expected result is that the data is successfully transmitted and becomes visible on the cloud dashboard.
- The fourth test case is AQI Calculation, where the gas sensor value is converted into an Air Quality Index. The expected result is that a valid AQI value is generated based on the sensor readings.

6.2 Test Procedure

- Simulated sensors in Wokwi
- Observed serial output
- Verified cloud updates
- Checked graph visualization

6.3 Performance Outcome

- System performed reliably
- Data transmission stable
- Real-time monitoring achieved

7 My learnings

During this internship, I gained:

- Practical knowledge of IoT systems
- Experience with ESP32 microcontroller
- Understanding of sensor interfacing
- Knowledge of cloud platforms
- Problem-solving and debugging skills

This experience will help me in future careers related to embedded systems and IoT.

8 Future work scope

- Implement real hardware instead of simulation
- Improve AQI accuracy using advanced sensors
- Develop mobile application for monitoring
- Use MQTT protocol for better efficiency
- Integrate Node-RED dashboard